

Healthcare SQL Project

MySQL Practice Project • 26 Questions • Basic, Intermediate & Advanced

Database

```
#health_care project 26 questions
use health_care;
#Tables: patients, doctors, appointments, medical_records
#Basic Level – 10 Questions
```

Basic Level

Q1. Display all records from the patients table;

```
create view patients_view as
  SELECT * FROM patients;
select*from patients_view;
```

Q2. Display the patient name, gender, and age.

```
create view patient_choosecolumn as
select patient_name,age,gender from patients;
select*from patient_choosecolumn;
```

Q3. Find patients whose age is greater than 60.

```
create view patients_view as
select *from patients
where age >60;
select *from patients_view;
```

Q4. Find all unique blood groups of patients.

```
create view dist_bloodgroup as
select distinct blood_group from patients;
select*from dist_bloodgroup;
```

Q5. Display doctors in descending order of experience.

```
create view doctorexperience AS
select* from doctors
order by experience_years desc;
select*from doctorexperience;
```

Q6. Display all completed appointments.

```
create view completedappoint as
select * from appointments
where appointment_status = "completed";
select*from completedappoint;
```

Q7. Find appointments where the consultation fee is greater than 800.

```
create view consultationfee as
select* from appointments
where consultation_fee >800;
select * from consultationfee;
```

Q8. Find the total number of patients.

```
create view totalpatient as
select count(patient_id) as totalpatient
from patients;
select*from totalpatient;
```

Q9. Find the average age of all patients.

```
create view allpatients as
select avg(age) as allpatients from patients;
select*from allpatients;
```

Q10. Find all unique doctor specializations

```
create view distspecialization as
select distinct specialization from doctors;
select * from distspecialization;
#Intermediate Level – 10 Questions
```

Intermediate Level

Q11. Display patient names with their appointment details using JOIN.

```
create view patientandappointmentdetails as
```

```

select p.patient_name,
a.appointment_id,
a.doctor_id,
a.appointment_date,
a.appointment_status,
a.consultation_fee from patients p
inner join appointments a
on p.patient_id=a.patient_id;
select*from patientandappointmentdetails;

```

Q12. Find the number of appointments for each doctor.

```

create view Each_doctor_appointments as
select doctor_id,
count(*) as total_appointment from appointments
group by doctor_id;
select*from Each_doctor_appointments;

```

Q13. Find the total consultation fee collected by each doctor.

```

create view collected_ech_doctors as
select d.doctor_id,
d.doctor_name ,
sum(a.consultation_fee) as total_consultation_fee
from doctors as d
inner join appointments as a
on d.doctor_id=a.doctor_id
group by d.doctor_id,d.doctor_name;
select* from collected_ech_doctors;

```

Q14. Find the average consultation fee for each appointment type.

```

create view avg_consultation_fee as
select
appointment_type,
avg(consultation_fee) as Avg_consultation_fee
from appointments
group by appointment_type;
select*from avg_consultation_fee;

```

Q15. Find patients who have more than 2 appointments.

```

create view more_than_2_appointment as
select patient_name,count(a.appointment_type) as more_than_2_appointment
from patients as p
inner join appointments as a
ON p.patient_id=a.patient_id
group by patient_name
having count(appointment_type) >2;
select*from more_than_2_appointment;

```

Q16. Find the top 5 doctors with the highest number of appointments.

```

create view top_five_doctor as
select d.doctor_name,d.doctor_id,
count(a.appointment_id) as top_five_doctor
from doctors as d
inner join appointments as a
on d.doctor_id=a.doctor_id
group by d.doctor_name,d.doctor_id
order by top_five_doctor desc
limit 5;
select* from top_five_doctor;

```

Q17. Find the number of completed appointments for each city.

```

create view total_completed_appointments as
SELECT p.city,COUNT(a.appointment_status) AS total_completed_appointments
FROM patients as p
inner join appointments as a
on p.patient_id=a.patient_id
WHERE appointment_status = 'Completed'
GROUP BY p.city;
select*from total_completed_appointments;

```

Q18. Find the total number of medical records for each diagnosis.

```

create view each_total_record as
select diagnosis,
count(*) as each_total_record from medical_records
group by diagnosis;
select*from each_total_record;

```

Q19. Find patients who have at least one medical record.

```

create view one_medical_recors as

```

```

select DISTINCT p.patient_id,p.patient_name,m.treatment
FROM patients AS p
INNER JOIN medical_records AS m
    ON p.patient_id = m.patient_id;
select*from one_medical_recors;

```

Q20. For each specialization, find the number of doctors and their average experience.

```

create view avg_exp as
select specialization,
avg(experience_years) as exp_year
from doctors
group by specialization;
select*from avg_exp;
#Advanced Level – 6 Questions

```

Advanced Level

Q21. Rank doctors based on their total consultation revenue using RANK().

```

create view doctor_fee_rank as
select d.doctor_id,d.doctor_name,
sum(a.consultation_fee) as total_revenue,
rank() over(order by sum(a.consultation_fee) desc) As doctor_rank
from doctors as d
inner join appointments as a
on d.doctor_id=a.doctor_id
GROUP BY d.doctor_id, d.doctor_name;
select*from doctor_fee_rank;

```

Q22. Find the latest medical record for each patient using ROW_NUMBER().

```

create view medical_record_each_patient as
select * from (
    select
        p.patient_id,
        p.patient_name,
        m.record_date,
        row_number() over (partition by p.patient_id order by m.record_date desc) as latest_patient
    from patients as p
    inner join medical_records as m
    on p.patient_id=m.patient_id) as y
where latest_patient =1;
select*from medical_record_each_patient;

```

Q23. Find the latest appointment for each patient.

```

create view appointment_latest as
select* from
(
    select p.patient_id,p.patient_name,a.appointment_date,
        row_number() over (partition by p.patient_id order by a.appointment_date desc) as latest_appointments
    from patients as p
    inner join appointments as a
    on p.patient_id=a.patient_id) as x
    where latest_appointments=1;
select*from appointment_latest;

```

Q24. Calculate the total appointments and consultation revenue for each month

```

create view each_month_revenue as
select
    monthname(appointment_date) as month,
    count(appointment_id) as total_appointments ,
    sum(consultation_fee) as total_revenue
from appointments
group by monthname(appointment_date)
order by month;
select*from each_month_revenue;

```

Q25. Calculate the running total of consultation revenue by appointment date.

```

create view RT_CONSULTATION AS
select
    appointment_id,
    appointment_date,
    consultation_fee, sum(consultation_fee)
over(order by appointment_date) as running_total_revenue
from appointments
order by appointment_date;
select*from RT_CONSULTATION ;

```

Q26. Find doctors whose appointment count is higher than the average appointment count of all doctors.

```

select

```

```

        d.doctor_id,
        d.doctor_name,
        count(a.appointment_type) AS c_app
from doctors AS d
inner join appointments AS a
    ON d.doctor_id = a.doctor_id
group by
    d.doctor_id,
    d.doctor_name
having count(a.appointment_type) > (
    select avg(doctor_count)
    FROM (
        SELECT COUNT(*) AS doctor_count
        FROM appointments
        GROUP BY doctor_id
    ) AS avg_doc
);

```